Assignment 3: Longitudinal Text Messaging Analysis

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Section 1: Environment Setup & Data Preparation

First, I loaded the `pastecs`, `tidyr`, and `dplyr` packages to perform data wrangling and descriptive statistics. Then, I imported `TextMessages.csv` from the project repository.

I converted `Group` into a labeled factor and the dataset into a tidy long format using `pivot_longer()` so that `Time` is a categorical variable with `Group`. Then, I exported this clean long dataset as `TextMessages_Clean_Long.csv` so my partner can import it directly to create the required faceted boxplots and bar charts.

```
# I loaded pastecs to compute detailed descriptive statistics using stat.desc()
library(pastecs)

# I loaded tidyr and dplyr to perform data wrangling and reshape the dataset
library(tidyr)
```

```
##
## Attaching package: 'tidyr'
```

```
## The following object is masked from 'package:pastecs':
##
##     extract
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:pastecs':
##
##     first, last
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
# I imported the shared dataset from the project repository into text_data
text_data <- read.csv("TextMessages.csv")

# I checked the initial wide structure and dimension of the raw data
str(text_data)
```

```
## 'data.frame':    50 obs. of  4 variables:
```

```
##  $ Group      : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ Baseline   : int  52 68 85 47 73 57 63 50 66 60 ...
##  $ Six_months : int  32 48 62 16 63 53 59 58 59 57 ...
##  $ Participant: int  1 2 3 4 5 6 7 8 9 10 ...
```

```
head(text_data)
```

```
##   Group Baseline Six_months Participant
## 1     1       52         32           1
## 2     1       68         48           2
## 3     1       85         62           3
## 4     1       47         16           4
## 5     1       73         63           5
## 6     1       57         53           6
```

```
# I converted Group into a labeled factor for clean categorical identification
text_data$Group <- factor(text_data$Group,
                          levels = c(1, 2),
                          labels = c("Group 1", "Group 2"))

# I reshaped Baseline and Six_months from wide into tidy long format
text_long <- text_data %>%
  pivot_longer(
    cols = c(Baseline, Six_months),
    names_to = "Time",
    values_to = "Text_Messages"
  ) %>%
  mutate(
    Time = factor(Time,
                  levels = c("Baseline", "Six_months"),
                  labels = c("Baseline", "6 Months"))
  )

# I verified the structure of the reshaped long data frame
str(text_long)
```

```
## tibble [100 x 4] (S3: tbl_df/tbl/data.frame)
##  $ Group        : Factor w/ 2 levels "Group 1","Group 2": 1 1 1 1 1 1 1 1 1 1 ...
##  $ Participant  : int [1:100] 1 1 2 2 3 3 4 4 5 5 ...
##  $ Time         : Factor w/ 2 levels "Baseline","6 Months": 1 2 1 2 1 2 1 2 1 2 ...
##  $ Text_Messages: int [1:100] 52 32 68 48 85 62 47 16 73 63 ...
```

```
head(text_long)
```

```
## # A tibble: 6 x 4
##   Group   Participant Time     Text_Messages
##   <fct>         <int> <fct>            <int>
## 1 Group 1           1 Baseline            52
## 2 Group 1           1 6 Months            32
## 3 Group 1           2 Baseline            68
## 4 Group 1           2 6 Months            48
## 5 Group 1           3 Baseline            85
## 6 Group 1           3 6 Months            62
```

```
# I saved the clean long data frame so my partner can use it for visual tasks
write.csv(text_long, "TextMessages_Clean_Long.csv", row.names = FALSE)
```

Section 2: Stratified Summary Statistics by Group and Time

I calculated summary statistics for `Text_Messages` stratified jointly by `Group` and `Time` using `by()` paired with `stat.desc()` from the `pastecs` package.

```
# I computed stratified summary statistics across Group and Time using stat.desc
stratified_summary <- by(
  text_long$Text_Messages,
  list(text_long$Group, text_long$Time),
  stat.desc,
  basic = TRUE
)

# I displayed the stratified summary statistics for Group 1 and Group 2 across both time points
stratified_summary
```

```
## : Group 1
## : Baseline
##      nbr.val     nbr.null       nbr.na          min          max        range
##    25.000000     0.000000     0.000000    47.000000    85.000000    38.000000
##          sum       median         mean      SE.mean CI.mean.0.95          var
##  1621.000000    64.000000    64.840000     2.135946     4.408377   114.056667
##      std.dev     coef.var
##    10.679732     0.164709
## ------------------------------------------------------------
## : Group 2
## : Baseline
##      nbr.val     nbr.null       nbr.na          min          max        range
##   25.0000000    0.0000000    0.0000000   46.0000000   89.0000000   43.0000000
##          sum       median         mean      SE.mean CI.mean.0.95          var
## 1640.0000000   65.0000000   65.6000000    2.1671794    4.4728385  117.4166667
##      std.dev     coef.var
##   10.8358971    0.1651814
## ------------------------------------------------------------
## : Group 1
## : 6 Months
##      nbr.val     nbr.null       nbr.na          min          max        range
##   25.0000000    0.0000000    0.0000000    9.0000000   78.0000000   69.0000000
##          sum       median         mean      SE.mean CI.mean.0.95          var
## 1324.0000000   58.0000000   52.9600000    3.2662313    6.7411700  266.7066667
##      std.dev     coef.var
##   16.3311563    0.3083678
## ------------------------------------------------------------
## : Group 2
## : 6 Months
##      nbr.val     nbr.null       nbr.na          min          max        range
##   25.0000000    0.0000000    0.0000000   46.0000000   79.0000000   33.0000000
##          sum       median         mean      SE.mean CI.mean.0.95          var
## 1546.0000000   62.0000000   61.8400000    1.8820910    3.8844450   88.5566667
##      std.dev     coef.var
##    9.4104552    0.1521742
```

Summary:

Group 1 Statistics: * **Baseline (n = 25):** Mean = 64.84, Median = 64.00, SD = 10.68, Variance = 114.06, Range = 38.00 (Minimum = 47.00, Maximum = 85.00), SE = 2.14. * **6 Months (n = 25):** Mean = 52.96, Median = 58.00, SD = 16.33, Variance = 266.71, Range = 69.00 (Minimum = 9.00, Maximum = 78.00), SE = 3.27.

Group 2 Statistics: * **Baseline (n = 25):** Mean = 65.60, Median = 65.00, SD = 10.84, Variance = 117.42, Range = 43.00 (Minimum = 46.00, Maximum = 89.00), SE = 2.17. * **6 Months (n = 25):** Mean = 61.84, Median = 62.00, SD = 9.41, Variance = 88.56, Range = 33.00 (Minimum = 46.00, Maximum = 79.00), SE = 1.88.

Inferences:

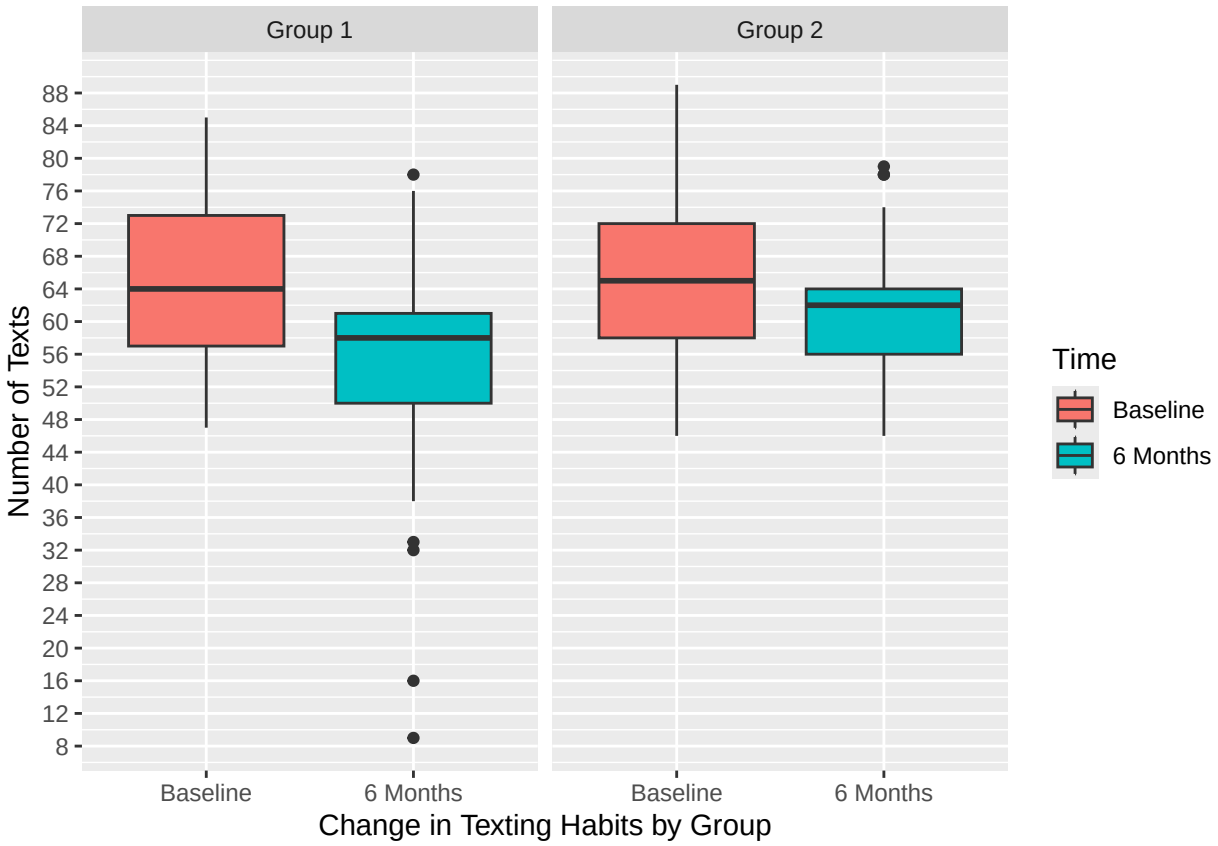
1. **Baseline:** At Baseline, both intervention groups exhibited nearly identical messaging rates (Group 1 Mean = 64.84 vs. Group 2 Mean = 65.60) and very similar variability (SD = 10.68 vs. 10.84), showing that the groups were comparable prior to intervention.
2. **Longitudinal Change:** Over the 6-month period, Group 1 demonstrated a substantial decrease in text messaging, dropping by an average of 11.88 messages (from 64.84 down to 52.96). In contrast, Group 2 showed a much smaller reduction of only 3.76 messages (from 65.60 down to 61.84).
3. **Variance:** While Group 1 experienced a greater reduction in text messages, its spread widened considerably at 6 months (SD increased from 10.68 to 16.33; range increased from 38 to 69). In contrast, Group 2 maintained a tight, consistent spread (SD decreased from 10.84 to 9.41; range decreased from 43 to 33).

Section 3: Boxplot Visualization by Group and Time

```r
#The ggplot2 package is first loaded to construct visualizations
library(ggplot2)

#To construct a boxplot that not only shows the change in texting habits over a
#period of six months, but also shows the difference between Group 1 and Group
#2's change in texting, a faceted approach is used.
boxplot_text <- text_long %>% ggplot(aes(Time, Text_Messages, fill=Time))+
  geom_boxplot()+
  labs(x="Change in Texting Habits by Group", y="Number of Texts")+
  scale_y_continuous(breaks=seq(from=0, to=90, by=4))+
  facet_wrap(Group~.)

#To view the faceted boxplot, the object is re-typed.
boxplot_text
```



From this, we are able to see that Group 2 had both the highest and lowest baseline texting scores, as well as the highest score after six months, while Group 1 had the lowest score after six months. Group 2 has a baseline score that appears to be approximately normally distributed, whereas Group 1 appears to have a slightly right skewed distribution. However, both follow-up scores are noticeably left skewed. Both groups had a noticeable decrease in the number of text messages sent after six months, with Group 1 appearing to have a greater decrease in median scores (from 64 to 58 versus 65 to 62). Althought Group 1 has a slightly lower median baseline score than Group 2, its interquartile range is slightly greater, suggesting a greater spread throughout the middle 50% of the data. Further, Group 1 had considerably low outliers in its six month scores, whereas Group 2 had multiple high outliers. Finally, both groups have a much smaller interquartile range at six months compared to their baseline scores, consistent with a smaller amount of variability among the middle 50% of follow-up scores.

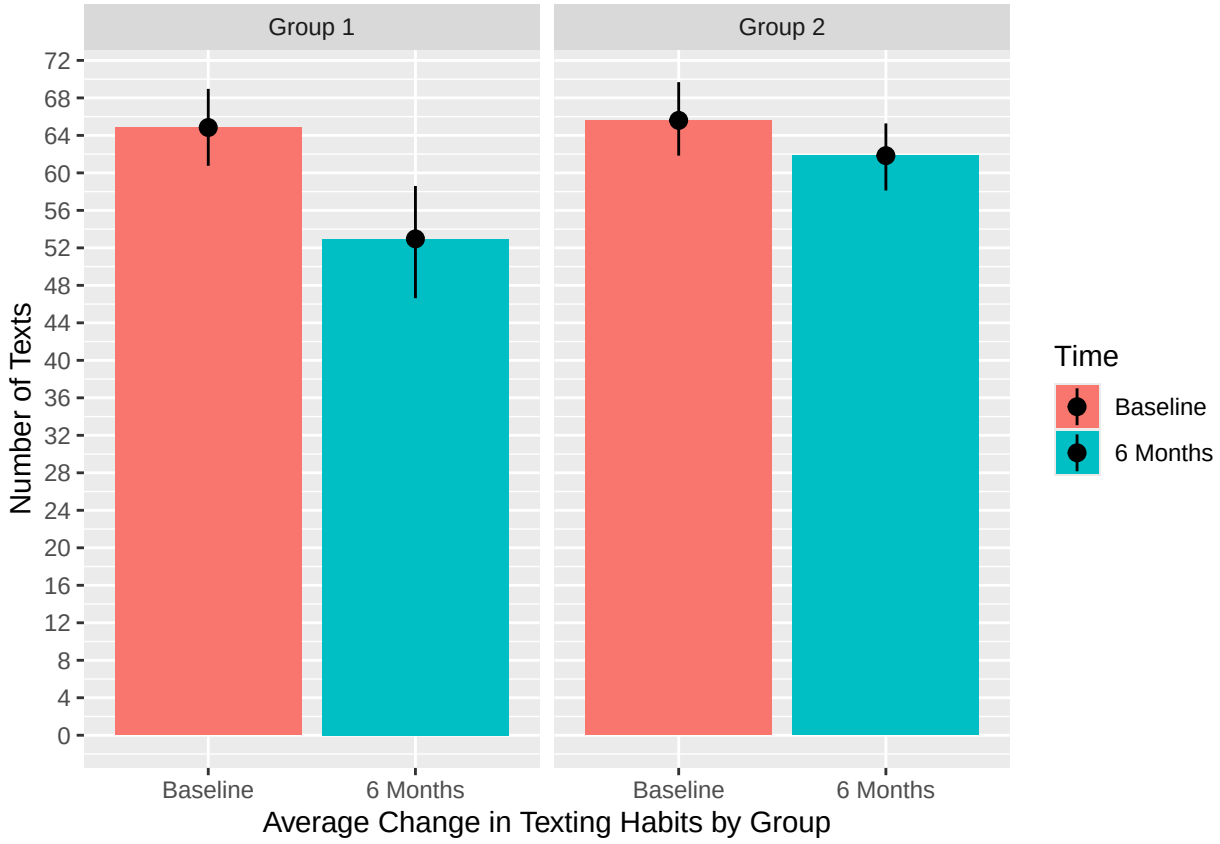
Section 4: Bar Chart Visualization by Group and Time

```
#The Hmisc package is loaded to construct errorbars that represent the 95%
#confidence interval of the mean.
library(Hmisc)

#Similarly, to construct a bar graph that displays a side-by-side comparison of
#the average baseline score and the average six month score for Groups 1 and 2,
#a faceted approach is used once more. An error bar is also included to depict
#the 95% confidence interval for each score's average.
barchart_text <- text_long %>% ggplot(aes(Time, Text_Messages, fill=Time))+
  stat_summary(fun=mean, geom="bar")+
```

```
  facet_wrap(Group~.)+
  scale_y_continuous(breaks=seq(from=0, to=100, by=4))+
  stat_summary(fun.data = mean_cl_boot, geom= "pointrange", colour="Black")+
  labs(x="Average Change in Texting Habits by Group", y="Number of Texts")

#To view the bar chart, the object is re-typed.
barchart_text
```



Beginning with the baseline scores, we are able to see that Group 1 has a slightly lower average score than Group 2, of approximately 64.84 versus 65.60, consistent with the previously examined summary statistics. Group 2 appears to have a slightly larger 95% confidence interval than Group 1. Given that both groups have the same sample size, this can be attributed to Group 1 having a lower standard deviation than Group 2, of 10.68 versus 10.84, respectively. Both groups exhibit a decrease in average texting scores after six months, with Group 1 not only having a smaller average follow-up score than Group 2 (of 52.96 versus 61.84), but also a much greater decrease in average scores of approximately 12 versus 4. Unlike the baseline scores, we can see that the six month scores have a larger difference between the two 95% confidence intervals, with Group 1 having a greater distribution of scores about the mean than Group 2. From this, we can conclude that Group 1 also had a greater change in variability than Group 2 after six months.